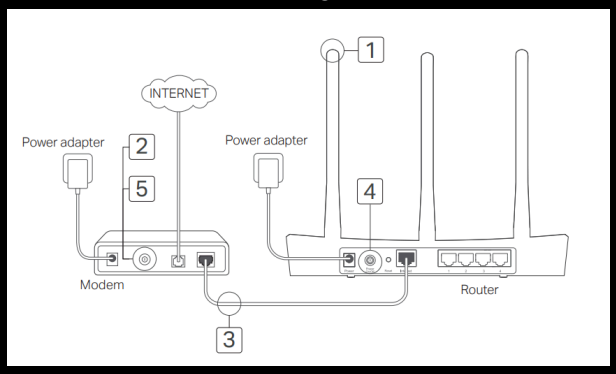
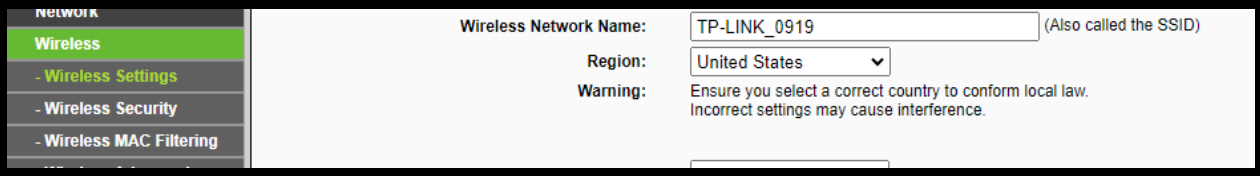
Wireless Networks

1. Recently you had a chance to connect your computer to a router. You then were able to access the router via a browser based interface by using the IP address of the router. You also connected another computer to the router and then pinged each computer from the other. Both computers were connected to the LAN ports of your router forming a wired LAN or local area network. The blue WAN port is what you would normally use to connect one router to another. This connects one network to another. That's the purpose of a router: to connect one network i.e. the one in your home to another i.e. the internet.
2. In your home network the router is usually connected to a modem before leaving your home and connecting to another router. The modem is used to convert the signal coming from outside your home into a form your home network can understand and then it converts it back to the signal the hardware outside your home can understand.

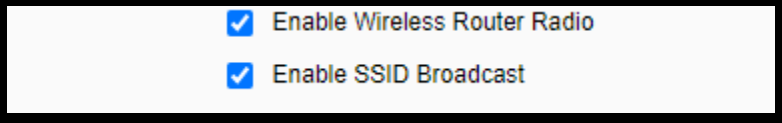


3. In these next few steps we will understand more about the settings in a router as well as how to use its wireless features to set up a wireless LAN.
4. Using a computer connected to the router's LAN port via an ethernet cable, access the router web admin interface using its IP address (usually 192.168.0.1). Using this interface answer the following questions and complete the exercises:
5. Wireless Network Name: If you select the **Wireless Settings** you should find an **SSID**. This is the name of the wireless connection:



What is the default name for your router's SSID (make sure to save it/update the settings)?
TP-link_EF08

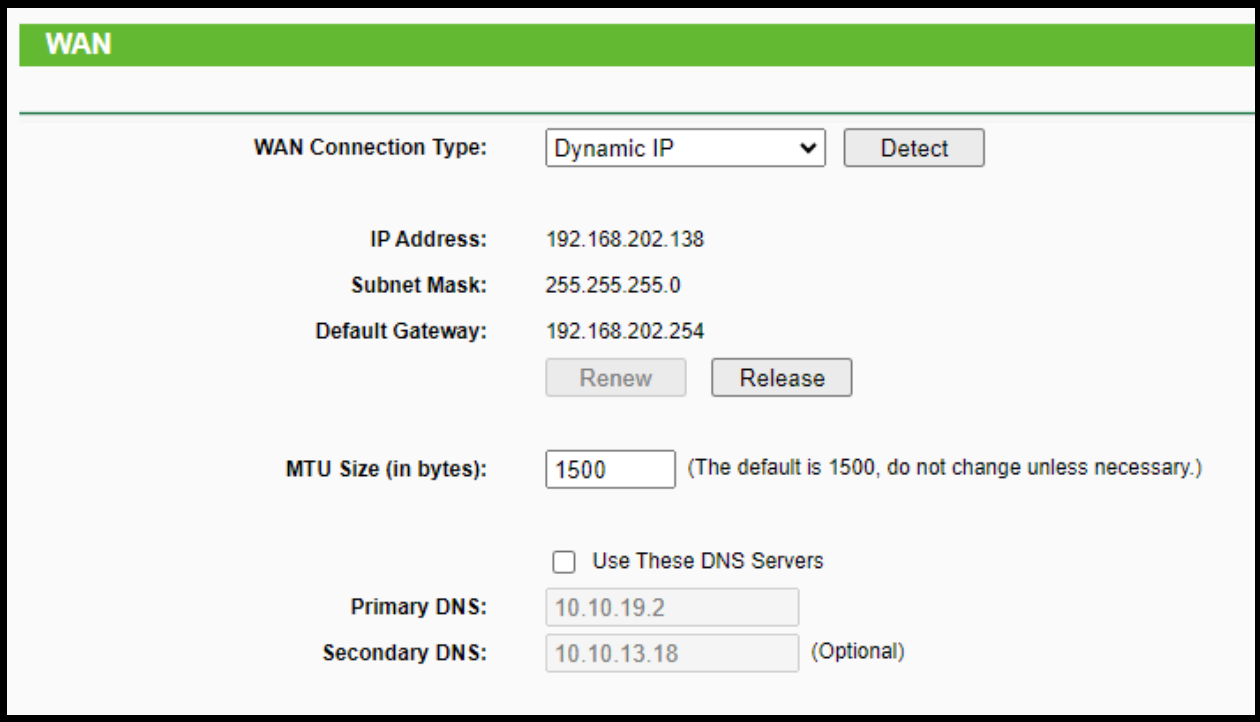
6. Change this to another name and look for it in your cell phone's wifi settings. If you don't see it you may have to restart the router after saving the settings.
7. Ask your teacher for a wifi dongle. Connect it to your computer via the USB port and look for the SSID of your router. Usually you will see other SSIDs. Why?
There are other ssid's of other routers in the classroom
8. Look for the SSID using the wifi access features of your cell phone. Can you see it?
yes
9. Uncheck the **Enable SSID Broadcast** option. Save and restart if necessary. Check your wifi settings on your cell phone. What do you notice and why does this happen?



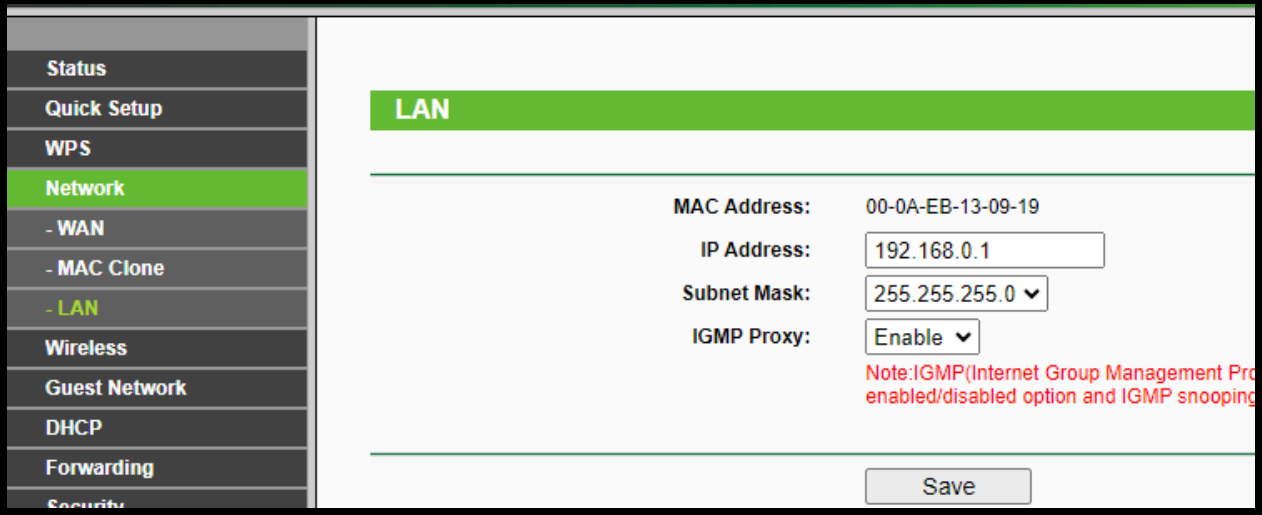
The wifi network is no longer visible in the list of networks but you could still connect by manually entering the name

10. What do you think the **Enable Wireless Router Radio** option is for?
It turns on and off the wireless capabilities and connections of the router

11. Make sure to enable **SSID broadcast** again.
12. Enter the **Wireless Security** tab and list the security options. Do some research and explain the differences between them.
    WPA
    WPA2
    WPA3
13. Enter the **Network→WAN** tab:

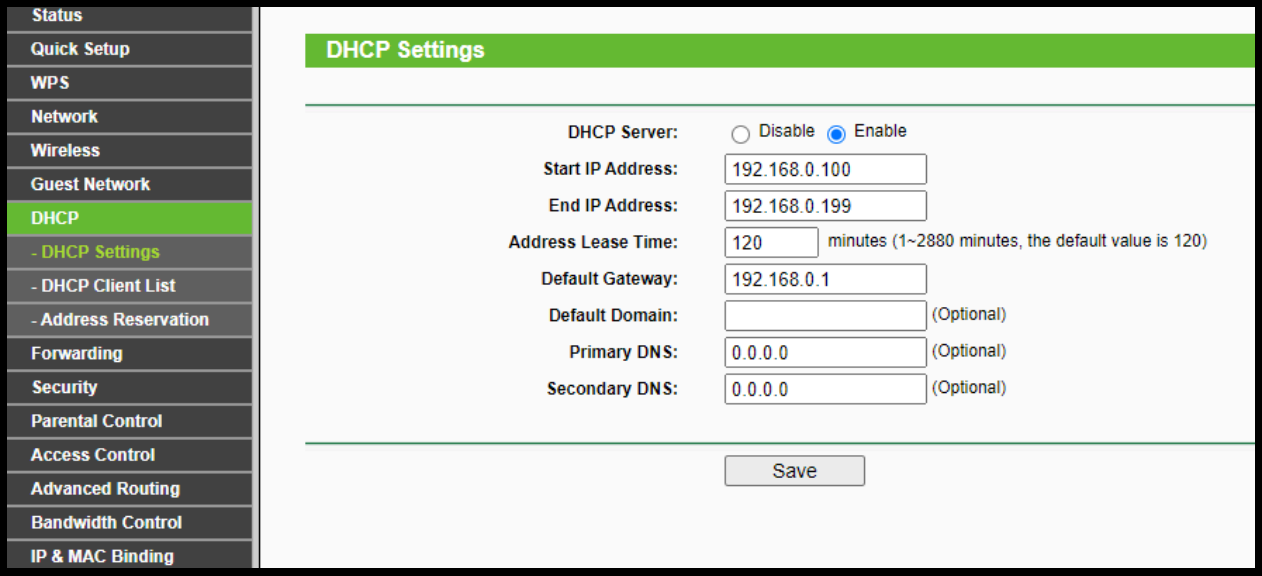


14. Explain what the **Dynamic IP** setting does?
    The dynamic IP setting makes it so your router tries getting an IP address from the ISP instead of setting one
15. What is the **IP Address**? What does this represent? Explain its value.
    The unique address given to a device that represents it within a network. It allows your router and ISP to differentiate between devices in a network
16. What is the subnet mask?
    A guideline for ip addresses
17. What does this represent?
    What parts of the IP address are the network address and what part is the host address
18. What is the **Default Gateway**?
    The IP address for the router
19. What does this represent?
    The IP address for the router
20. What are the **Primary** and **Secondary DNS IP** addresses and what do they represent?
    They are the ip addresses for the 2 DNS servers, one of which is read/write and does the primary work of converting website names to IP addresses, while the second one is read/only and helps with repetition and redundancy
21. Enter the **Network LAN** tab. What does the **MAC Address** represent?



The mac address is the physical address of the LAN ports, as seen from the LAN. The value cannot be changed

22. What does the **IP Address** represent?
    The IP address of the router
23. What does the **Subnet Mask** represent?
    The subnet mask for the network, shows what part of the IP is network and host address
24. Click on the **DHCP** tab.



25. What is the role of **DHCP**?
    It is a protocol that assigns unique IP addresses to each device in the network
26. What does selecting **Disable** do?
    It prevents the router from automatically assigning IP addresses to the other devices in the network
27. What is the purpose of setting the **Start** and **End IP Addresses**?
    It tells you the range of IP addresses that can be assigned to the connected devices
28. What is the purpose of the **Address Lease Time** setting?
    It determines how long the routers DHCP server will reserve an IP address for a device
29. Click on the **DHCP Client List** tab. What do you notice? Now check the IP address of your phone (in android you should be able to go to **About Phone** and find it there) and relate your findings to the DHCP Client List.
    My phone's IP address is on the DHCP client list because my phone is connected wirelessly
30. Go to the Wireless Security settings and change your wireless password. Save this. Disconnect the ethernet cable from your computer and attempt to connect to the router wirelessly. Work with another group or find another computer with a wifi dongle and

attempt to connect to the same router wirelessly. If both connected then you have successfully created a wireless LAN.

31. Now that you have successfully joined your local network through a wireless connection your next task is to find a local multiplayer game for the android and/or iphone and set it up so that you can play someone else over the LAN you just made. Here's a reference site for a list of possible games to test out: https://www.makeuseof.com/tag/best-local-multiplayer-games-android-iphone/
32. Demonstrate your game to me. A bonus mark will be assigned for it.
    done